PROGRESS REPORT

Evolutionary conservation, positive selection and structural classification: an integrative analysis of EDAR

Tran, Nhat Linh

1. Key Findings

1.1 Species Selection

Alignment covers ~20 vertebrates selected to cover phenotypic differences of ectodermal traits:

- Primates: closest relatives
- Hairy mammals: dense fur -> expects EDAR to be active
- Hairless/less hair mammals: reduced/no hair -> expected EDAR to be weaker
- Birds: feathers instead of hair
- Reptiles: scales
- Fish: outgroup

1.2 Multiple Sequence Alignment:

It shows three major functional regions that are conserved and identified with domain annotations:

- TNFR-Cys domain (CRD 1-3): cysteine residues highly conserved, loop regions more variable
- Transmembrane domain: conserved with many hydrophobic residues
- Death domain: strong conservation

Fish show increased divergence at the N-terminus, which correlates with their different functional context

1.3 Analysis of Position 370

Amino acid was identified in all 24 species, revealing this pattern:

Taxonomic Group	AA at 370	Interpretation
Primates	Val (V)	Ancestral state in mammals
Hairy mammals	Val (V)	Val in dense-fur species
Hairless mammals	Val (V)	Val also in hairless species
Birds	Ile (I)	substitution (both hydrophobic)
Reptiles	Ile (I)	Same as birds, shared ancestral substitution
Fish	Val / Ile	Higher divergence

Homo sapiens EAS	Ala (A)	Alanine substitution affects the hydrophobic core of the Death Domain.

Position 370 is moderately conserved, which can be the reason for the positive selection of V370A and having a functional effect but is not lethal. (Method to calculate score: Rate4Site~0.1467 || UCSC Table Browser)

---

2. Key Findings

2.1 Conservation Profile

With Rate4Site, it calculated the score for each amino acid and then I looked at the conservation profile plot (protein level), which is consistent with the sequence alignment.

- CRDs: show variable and conserved patterns
- Transmembrane domain: shows mostly negative scores (strongly conserved)
- Death domain: shows also predominantly negative scores

I am currently working on plotting Shannon Entropy from my multiple sequence alignment.

2.2 Working on Hypothesis

Cross-species domain analysis reveals that position 370 is located within the death domain, which is conserved across analyzed vertebrates.

The death domain of EDAR is conserved across vertebrates and functionally indispensable. Nevertheless, V370A demonstrates that positive selection can act within this conserved domain. It is hypothesized that the Val to Ala substitution at position 370 is structurally tolerable and functionally relevant by modulating EDAR signaling strength and is positively selected in East Asian populations due to a phenotypic advantage in ectodermal development without disrupting the core function of the death domain.